

The Logic of Economic Choice

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Required readings

The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 1** — the principles of economic choice.
- **Chapter 2** — how economists think, and the production possibility frontier.

Read both chapters alongside these slides.

1. **Scarcity and trade-offs** — why choice is forced on us.
2. **Opportunity cost and sunk cost** — what a choice really costs.
3. **The margin, rationality, incentives** — how choices are made.
4. **Trade, markets, government** — how choices fit together.
5. **The economist as scientist** — models and *ceteris paribus*.
6. **The production possibility frontier** — the workhorse diagram.
7. **Increasing opportunity cost** — why frontiers usually bend.
8. **Efficiency, unemployment, growth** — reading the frontier.
9. **Facts and values** — what evidence can and cannot settle.

PART 1

Scarcity and Trade-offs

People want more than they can have

Scarcity. The condition in which society's wants exceed the resources available to satisfy them.

Economics (Robbins, 1932). Economics is the science which studies human behavior as a relationship between ends and scarce means which have alternative uses.

Scarcity is not poverty. A country with \$80,000 per person faces it just as surely as one with \$2,000. **At every income level, choosing one thing means forgoing another.**

What that definition is actually saying

Four words are doing the work.

- **Ends** — things people are trying to achieve. Economics takes them as given; it does not rank them.
- **Means** — the resources available: time, labor, land, capital.
- **Scarce** — there are not enough means to serve every end.
- **Alternative uses** — the same means could serve a *different* end.

Economics is the science which studies human behavior as a relationship between ends and scarce means which have alternative uses.

Microeconomics narrows the lens

Microeconomics studies the decisions of **individual households and firms**, and the markets in which they interact.

- How does a consumer decide what to buy?
- How does a firm set its price?
- How does a labor market determine wages?

Every one has the same structure: an agent faces a constraint and makes a choice.

Principle 1 — people face trade-offs

Every choice has a cost.

- Studying economics for an hour means not studying psychology, not working a shift, not sleeping.
- A government that allocates more to defense allocates less to healthcare, infrastructure, or tax cuts.
- Engineers assigned to product A cannot simultaneously build product B.

The point is not that trade-offs are bad. **The point is that they exist, and ignoring them leads to poor decisions.**

Two different questions

Allocation. How scarce resources are assigned among competing uses. Determines **how much** gets produced — the size of the pie.

Distribution. Who ends up with the output. Determines **who gets which slice**.

In competitive markets, earnings partly reflect **factor productivity**; ownership, institutions, and policy matter too.

These are separate questions. We are mostly concerned with the first.

So what does the allocating?

Scarcity forces choices. Something has to guide them.

A resource could be *directed* — by a planner, a ministry, a committee. Or it could be allocated by **prices**.

In a market economy, the price system guides allocation — without a central planner.

How that is even possible is the question the rest of this lecture answers.

Active Learning 1

For each scenario: **what exactly is given up**, and is this an **allocation** question, a **distribution** question, or both?

1. A city rezones industrial land for housing.
2. A progressive income tax funds transfers to low-income households.
3. You take economics instead of a psychology elective this term.

On paper. Two minutes. One of these is not as clean as it looks.

PART 2

Opportunity Cost and Sunk Cost

Principle 2 — the cost of something is what you give up to get it

Opportunity cost. The value of the best alternative forgone when a choice is made.

This is the central cost concept in economics, and it differs from *money spent* in two directions:

- it **counts** real costs that never appear on an invoice;
- it **drops** money that is already gone.

The cost of a year of college

Cost	Amount	Appears on a bill?
Tuition and fees	\$32,000	yes
Foregone earnings ($\$19/\text{hr} \times 2,000 \text{ hrs}$)	\$38,000	no

In this example, the largest cost never appears on a bill.

Both are real costs of the same decision.

Sunk cost. A cost that has already been incurred and cannot be recovered, regardless of what decision is made now.

Sunk costs should not influence current decisions.

Whatever you choose next, that money is spent either way.

The concert ticket

You paid \$150 for a concert ticket. On the evening of the show, you fall ill.

The wrong reasoning: “I’ve already spent the money, so I’d better go.”

The right question: does attending, ill, yield more value than resting?

The \$150 is gone whether you attend or stay home. It is irrelevant to the decision.

What counts and what does not

Ahead of you — the best alternative you would give up: *count it*. That is the opportunity cost.

Behind you — money already spent, unrecoverable: *ignore it*. That is sunk.

Only costs a decision can still change belong in the decision.

The habit to build

Thinking in opportunity cost, rather than money spent, is one of the most useful habits economics can give you.

The discipline is a single question, asked before every decision: **what did I give up?**
— not *what did I pay?*

The two give different answers surprisingly often, and the second is almost always the one that matters.

In Part 6 the slope of the frontier turns out to *be* an opportunity cost.

Active Learning 2

You have one free evening. Three options:

- work a shift paying **\$55**;
- attend a review session you value at **\$40**;
- stream a movie you value at **\$15**.

You choose the review session.

1. What is the opportunity cost of your choice?
2. Yesterday you prepaid a nonrefundable \$20 to rent that movie. Does that change your answer? Why or why not?

On paper. Three minutes. The *why* is the marks.

PART 3

The Margin, Rationality, Incentives

Principle 3 — decisions happen at the margin

Most economic decisions are not all-or-nothing.

- A firm does not choose between producing nothing and producing everything — it chooses whether to produce *one more unit*.
- A student chooses whether to study *one more hour*.
- A hospital decides whether the *next* diagnostic test is worth its cost.

Marginal analysis. The comparison of the additional benefit (marginal benefit, MB) and additional cost (marginal cost, MC) of one more unit of an activity.

Take the action if $MB > MC$; $MB = MC$ is the boundary.

For an incremental choice, rational analysis compares the next unit's benefit with its cost.

The empty seat

A flight from Boston to Los Angeles costs the airline \$100,000 to operate, with 200 seats.

Average cost per seat	$\$100,000 \div 200 = \500
Standby passenger offers	\$300
Marginal cost of carrying her	a drink, a little fuel — near zero

Should the airline refuse? **No.** If the seat would fly empty, \$300 beats the marginal cost easily.

Average cost is irrelevant to the marginal decision.

Principle 4 — people respond to incentives

Incentive. Anything that alters the benefits or costs of an action, thereby influencing behavior.

- Gasoline prices rise → people drive less, buy smaller cars, relocate closer to work.
- A solar subsidy appears → homeowners install panels.
- Merit scholarships are offered → more students apply.

Change the relative costs or benefits, and behavior shifts — predictably.

The unintended margin — Peltzman (1975)

Federal automobile-safety rules made accidents **more survivable**. Done?

Peltzman found accident patterns **consistent with an offsetting response**: some safety gains were offset by more risk-taking.

- Overall highway deaths: not reduced.
- Time-series evidence suggested gains for occupants were offset by nonoccupant deaths and nonfatal accidents.

Sam Peltzman, "The Effects of Automobile Safety Regulation," *Journal of Political Economy* 83(4), 1975.

Trace every margin

A policy changes incentives on the margin it targets — **and on every other margin people can adjust.**

Good policy analysis traces incentives through all margins of adjustment, not merely the intended one.

This is hard. **That is why economics is useful.**

The incentive that runs an economy

Most incentives are deliberate — someone designs them. A tax, a subsidy, a seatbelt law.

The market price system is an incentive mechanism no central planner designed.

A price is not merely a number attached to a good. It makes some choices attractive and others less so — and it does this to *everyone at once*.

One number, two instructions, pointing opposite ways. A higher price tells buyers to economize and sellers to produce more. Nobody has to be told which group they are in.

Which price? Relative, not absolute

Suppose coffee doubles and nothing else changes. Coffee is now expensive *relative to* tea, to lunch, to everything. **So people reallocate.**

In a frictionless thought experiment, suppose *every nominal amount* doubles together — prices, wages, assets, and obligations.

Nothing is expensive relative to anything. No reason to reallocate at all.

Every nominal number changed; relative prices did not. Relative prices guide reallocation.

A town replaces its flat monthly trash fee with a charge of **\$2 per bag** collected at the curb.

1. **Predict first:** what behavior is the charge meant to change?
2. List **two other margins** on which households can respond.
3. Name **one response** that could leave the town worse off than the flat fee did.

On paper. Three minutes. No arithmetic — the reasoning is the marks.

PART 4

Trade, Markets, Government

Principle 5 — trade makes people better off

If both parties did not expect to gain, **the transaction would not happen**. No altruism required.

Trade creates value through **specialization**: a surgeon and a plumber each do better specializing and trading services than attempting both jobs.

Trade is not a zero-sum game.

Comparative advantage can create gains **even when one side is better at everything**.

Principle 6 — markets coordinate economic activity

Market economy. An economic system in which resources are allocated through the decentralized decisions of households and firms, guided by prices.

Every society must answer three questions: **what** to produce, **how** to produce it, and **who** gets the output.

In a market economy, decentralized choices guided by prices answer them — without a central planner.

Nobody knows how to make a pencil

Adam Smith, 1776: individuals pursuing self-interest are led, as if by an invisible hand, to outcomes that serve society. The mechanism is the **price system**.

And here is the harder claim. No single person knows how to make a pencil — the timber, the graphite, the lacquer, the brass. That knowledge is scattered across thousands of people who have never met and never will.

Prices coordinate knowledge that nobody holds in full.

Hayek (1945) called this the knowledge problem.

Principle 7 — government can sometimes improve market outcomes

First, markets need what they cannot supply themselves: **property rights** and **contract enforcement**.

Market failure. A market, left to itself, fails to allocate resources efficiently. Two recurring sources: **externalities** and **market power**.

But *can* improve does not mean *will* improve — policy is written by politicians answering to political incentives.

Never ask whether the market outcome is perfect. Ask whether a *feasible* intervention does better, counting its own costs.

Active Learning 4

Three headlines. For each: which principle — **5 (trade)**, **6 (prices coordinate)**, or **7 (market failure)** — is at work, and why?

1. “After a freeze ruins the citrus crop, orange prices jump; shoppers switch to apples and growers plant new groves.”
2. “A carpenter and an electrician renovate each other’s houses.”
3. “Upstream factory’s runoff closes the town beach; the council debates a discharge rule.”

On paper. Three minutes. The *why* is the marks.

PART 5

The Economist as Scientist

The method: observe, theorize, test, revise

The same loop as physics:

1. **Observe:** countries with fast-growing money supplies tend to have high inflation.
2. **Theorize:** money growth drives prices up.
3. **Test:** collect data across countries and decades.
4. **Revise** when the predictions fail.

Formulate a hypothesis, derive testable implications, let the data arbitrate.

No laboratory — natural experiments instead

Economists cannot double a country's money supply “to see what happens.”

Instead: naturally occurring variation. A war disrupts oil supplies; a government rewrites its tax code; a pandemic shuts an economy.

Astronomy and evolutionary biology face the same constraint.

The discipline is in the method, not the laboratory.

Assumptions and simplification

A physicist studies a falling marble *assuming a vacuum* — false, but harmless for a dense object.

Economists simplify the same way: study trade in a world of **two countries and two goods**, not 200 and millions.

The art is choosing the right simplification for the question.

“All models are wrong, but some are useful.” — George Box

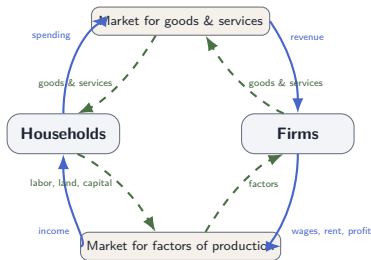
Ceteris paribus. “Other things being equal.” The assumption that all variables other than the one under study are held constant.

“A rise in the price of coffee reduces the quantity people buy, *ceteris paribus*” — holding income, tastes, and the price of tea constant.

Not a claim about the world — a device for clear thinking, one variable at a time.

Model 1 — the circular-flow diagram

Circular-flow diagram. A visual model of the economy showing the flows of goods, factors of production, and money between households and firms through two markets.



Three views of the same activity

Firms produce output (**production**); they pay households for factors (**income**); in this simple model households spend it on goods (**expenditure**).

Production, income, and expenditure are three measurements of the same flow. They must be equal.

The model omits government, banks, and other countries — **deliberate simplifications**, not oversights.

Active Learning 5

A coffee-shop owner tells you:

*“I raised my price 10% last month and sold more cups than ever —
so higher prices don’t reduce my sales.”*

1. What assumption does her argument ignore?
2. Name **two** things that might plausibly have changed in the same month.

On paper. Two minutes. The *why* is the marks.

PART 6

The Production Possibilities Frontier

Model 2 — the workhorse diagram

Production possibility frontier (PPF). The maximum combinations of two goods an economy can produce when it uses its resources fully and efficiently, given its technology.

One picture compresses three ideas: **scarcity** (a boundary exists), **productive efficiency** (points on it), **opportunity cost** (its slope).

Our economy makes **bicycles and jackets** with labor alone: **6,000 hours**; a bicycle takes **6 hours**, a jacket **2 hours**.

Step 1 — write the equation

Hours spent on bicycles plus hours spent on jackets equal total hours:

$$6B + 2J = 6000$$

Solve for J to put jackets on the vertical axis:

$$J = 3000 - 3B$$

Step 2 — plug the numbers in

$J = 3000 - 3B$		
Bicycles B	Substitute	Jackets J
0	$3000 - 3(0)$	3000
200	$3000 - 3(200)$	2400
400	$3000 - 3(400)$	1800
600	$3000 - 3(600)$	1200
800	$3000 - 3(800)$	600
1000	$3000 - 3(1000)$	0

The ends: **3,000 jackets** if no bicycles are made; **1,000 bicycles** if no jackets are.

Step 3 — draw the graph



Dots from the table, line from the function — coinciding. The shaded area is everything the economy *can* produce.

What the slope means

The slope is -3 : each extra bicycle costs **3 jackets**.

The slope of the frontier is the opportunity cost.

It came straight from the labor data:

$$\frac{6 \text{ hours per bicycle}}{2 \text{ hours per jacket}} = 3$$

Constant along the whole line — **that** is why the line is straight.

Use it — two routes to one answer

The economy makes 400 bicycles and wants 100 more.

Route 1, the equation: J falls from $3000 - 3(400) = 1800$ to $3000 - 3(500) = 1500$.

Route 2, the slope: $100 \text{ bicycles} \times 3 \text{ jackets each} = 300 \text{ jackets}$.

Both routes: 300 jackets given up. If they had disagreed, we would know to go looking.

The same slope, read the other way

Each bicycle costs 3 jackets. So each jacket costs

$$\frac{2 \text{ hours per jacket}}{6 \text{ hours per bicycle}} = \frac{1}{3} \text{ bicycle}$$

The two opportunity costs are **reciprocals**.

If bicycles are dear in jackets, jackets must be cheap in bicycles.

Build a frontier yourself.

An economy has **3,600 hours** of labor. A kayak takes **12 hours**; a paddle takes **3 hours**.

1. Write the labor constraint, then solve for paddles P as a function of kayaks K .
2. What is the maximum number of kayaks? Of paddles?
3. State the slope in words. Then **check it** against the ratio of labor hours — if the two disagree, something is wrong.

On paper. Four minutes.

PART 7

Increasing Opportunity Cost

Straight lines are a special case

The bicycle–jacket frontier was straight because **every labor hour was interchangeable** — each had the same relative productivity.

Real economies are not like that.

Resources differ. The land and the hands best at growing **food** are not the ones best at making **clothing**.

A bent frontier is still an equation

An economy produces **food** and **clothing**. Its frontier:

$$C = 50 - 0.02 F^2$$

Treat it exactly like the last one: **tabulate it, then plot it.**

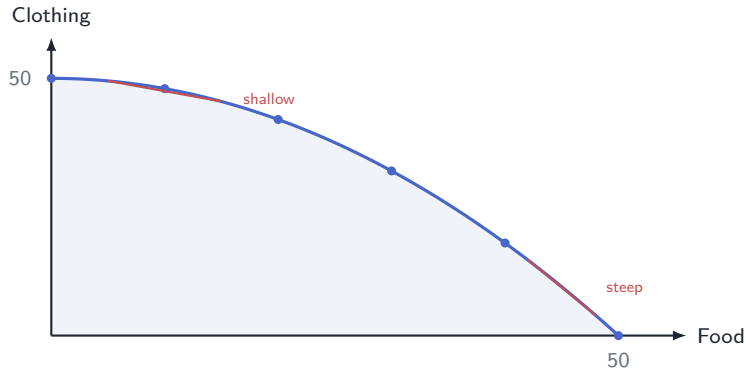
Watch one thing: **the cost of each extra 10 units of food.**

Plug the numbers in

Food F	$C = 50 - 0.02F^2$	Clothing C	Cost of the last 10 food
0	$50 - 0.02(0)$	50	—
10	$50 - 0.02(100)$	48	2
20	$50 - 0.02(400)$	42	6
30	$50 - 0.02(900)$	32	10
40	$50 - 0.02(1600)$	18	14
50	$50 - 0.02(2500)$	0	18

2, 6, 10, 14, 18. Each extra 10 food costs more clothing than the last.

The bowed frontier



The frontier bows outward. Its slope steepens as you move right.

Why frontiers bow

Start with all resources in clothing. Move some into food and you move the resources *best suited to food* first — **cheap**.

Keep going and you must pull in resources that were *genuinely good at clothing* — **expensive**.

Increasing opportunity cost is what specialized resources look like in a picture.

Two shapes, one rule

	Straight frontier	Bowed frontier
Opportunity cost	constant	increasing
Resources are	interchangeable	specialized
Which case	special	the normal one

Either way, the slope of the frontier is the opportunity cost.

Active Learning 7

An economy's frontier is $C = 90 - 0.1 F^2$.

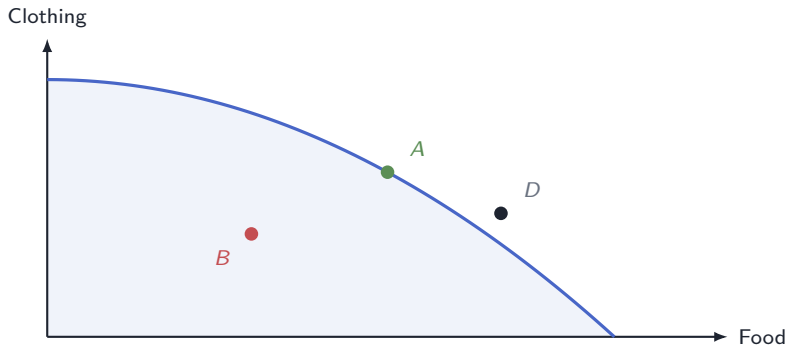
1. **Before computing anything:** will opportunity cost be constant or increasing?
How can you tell from the equation alone?
2. Compute C at $F = 0, 10, 20, 30$, showing the substitution.
3. What is the maximum food the economy can produce?
4. Compute the cost, in clothing, of each successive 10 units of food.

On paper. Four minutes.

PART 8

Efficiency, Unemployment, Growth

Three kinds of point



A on the frontier — efficient. **B inside** — inefficient. **D outside** — unattainable.

Points inside the frontier

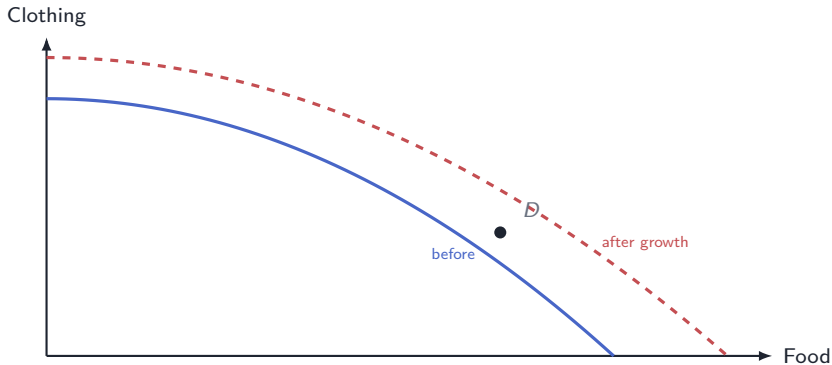
At B the economy could have **more food and more clothing at the same time**.
Nothing must be given up.

What puts an economy inside?

- **Unemployment** — workers willing to work, not working.
- Idle factories or land.
- Resources in the wrong use.

In this static model, some moves toward the frontier increase output without reducing the other good.

Growth — the frontier itself moves



More resources or better technology shifts the whole frontier outward. D **was unattainable; now it is inside.**

What moves it

- **More resources** — a larger workforce, more capital.
- **Better technology.**
- **Better institutions.**

Shifts can be **lopsided**: a food-only technology pushes the food end out more than the clothing end.

Ask not just *whether* the frontier moves, but *what shape* the move has.

Growth does not repeal scarcity

An outward shift means the economy *can* have more of both goods.

It does **not** eliminate scarcity: **trade-offs persist on the new frontier.**

Growth moves the boundary. It does not abolish it.

This diagram is the workhorse of the whole subject.

Active Learning 8

For each event: does the economy move **along** its frontier, is it pushed **inside**, or does the **frontier itself shift**? For shifts — what shape?

1. Clothing factories automate; output per worker in clothing doubles.
2. A strike idles a tenth of the workforce.
3. Resources are moved out of clothing into food.
4. New irrigation doubles food output per acre.

On paper. Two minutes. Say which — and why.

PART 9

Facts and Values

Two kinds of claim

Positive statement. A claim about how the world *is*. In principle, it can be confirmed or refuted by evidence.

Normative statement. A claim about how the world *ought to be*. It depends on values as well as facts.

The distinction decides **what evidence can settle**.

Two claims about the minimum wage

“A higher minimum wage reduces employment among teenagers.”

Positive. Testable: compare states, study changes over time. It may be true or false
— **data can arbitrate.**

“The government should raise the minimum wage.”

Normative. Even given the employment facts, it needs a *judgment*: how to weigh job losses against income gains. **Data can inform it, not settle it.**

Could evidence, in principle, prove this claim wrong?

Yes → positive. Depends on “should,” “ought,” “fair,” “just” → normative.

Beware smuggling. “The minimum wage destroys jobs” *sounds* scientific — but “...therefore it should not exist” does not follow without a value judgment.

And beware **“too”**: “*unemployment is too high*” imports a standard without stating it.

Correlation is not causation

In beach towns, **sunscreen sales** and **jellyfish stings** rise and fall together, month by month.

Does sunscreen attract jellyfish? **No.** A third variable — the summer crowds — drives both.

Omitted variable. A relevant factor left out of an analysis that affects the outcome and is correlated with the explanatory variable.

Sometimes the arrow points backward

Cities with **more police officers** have **higher crime rates**.

Does policing cause crime? Almost certainly not:

Crime affects police staffing, while policing may affect crime. Raw correlation entangles both directions.

Reverse causality — the second pervasive threat.

Identification — earning a causal claim

How economists address both threats:

- **Randomized controlled trials**, where feasible.
- **Natural experiments** — an event or rule creates plausibly comparable treated and untreated groups.
- **Instrumental variables** — a name worth recognizing, not a technique used here.

Be suspicious of causal claims that rest on raw correlation.

Why economists disagree

Scientific disagreement — different estimates of an effect's *sign or magnitude*. How does a 10% minimum-wage rise affect teenage employment? **Resolvable**, in principle, by better data.

Value disagreement — same facts, different *weights*. Two economists can agree on exactly what a policy does to allocation *and* to distribution, and still differ on how much each matters. **Not resolvable by evidence alone.**

Some disagreements yield to better evidence; disagreements about values do not.

Mark each claim **positive** or **normative**, with one line of justification:

1. "Raising the state gasoline tax by 50 cents would reduce driving by about 4 percent."
2. "The gasoline tax should be raised to fund public transit."
3. "Unemployment in this state is too high."
4. "Rent control reduces the number of apartments offered for rent."

On paper. Two minutes. The justification is the marks.

Every choice has an opportunity cost.

Scarcity forces choices; marginal choices compare the next unit's benefit with its cost; and the frontier's slope shows the production opportunity cost between its two goods. Economic claims must survive the evidence.

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